

Cloud Adoption & Migration : Experiment - 10

Aim:

To study and analyze secure connectivity mechanisms such as Virtual Private Networks (VPN) and VPC peering across multiple cloud environments, and to understand their role in enabling hybrid and multi-cloud architectures.

Theory:

In modern cloud computing, organizations often deploy applications across multiple cloud environments and on-premises systems. Secure communication between these isolated networks is essential for maintaining data integrity and performance. Technologies such as Virtual Private Networks (VPN) and VPC/VNet peering enable private and secure connectivity. These mechanisms help in building hybrid and multi-cloud architectures. This experiment focuses on studying and comparing these connectivity solutions.

Organizations operating in hybrid or multi-cloud environments require secure and reliable connectivity between isolated cloud networks. VPN and VPC peering enable private communication while maintaining security and performance.

VPC/VNet Architecture:

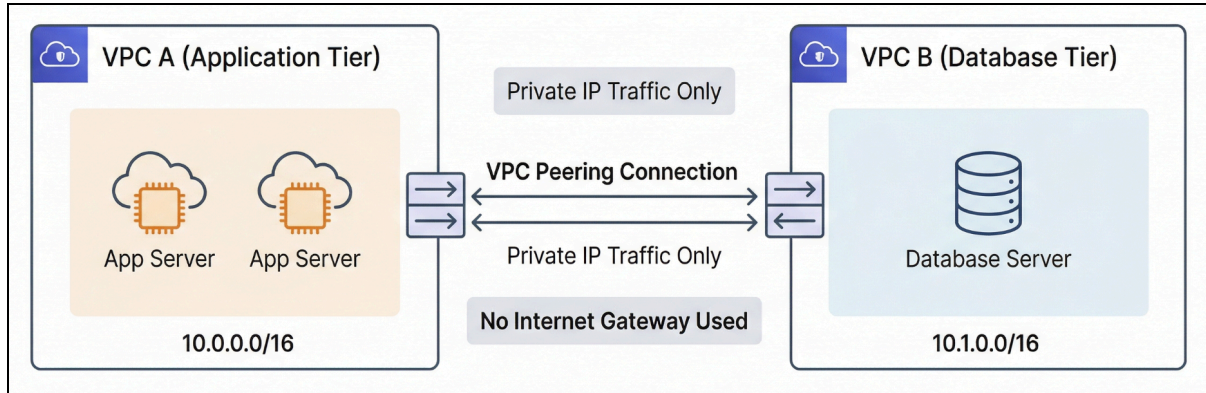
Components:

- VPC/VNet: A logically isolated virtual network within a cloud provider where resources are deployed with defined IP ranges.
- Subnets: Logical subdivisions of a VPC/VNet used to organize and segment resources based on functionality and security requirements.
- Route Tables: Define rules that control how network traffic is directed between subnets, gateways, and external networks.
- Internet Gateway: Enables communication between the VPC and the public internet for inbound and outbound traffic.
- NAT Gateway: Allows instances in private subnets to access the internet securely without exposing them to inbound traffic.
- Security Groups / NSG: Act as virtual firewalls that control inbound and outbound traffic at the instance or subnet level.

VPC Peering Concept:

VPC Peering connects two virtual networks within the same cloud provider using private IP addresses. It allows secure communication without using the public internet. It requires non-overlapping IP ranges and proper routing configuration. VPC peering is non-transitive and provides low latency.

- Traffic between peered VPCs flows through the cloud provider's internal backbone network, ensuring high speed and reliability.
- It does not require a gateway or VPN, making it a simple and cost-effective solution.
- Security groups and network access rules must be configured properly to allow communication between VPCs.
- VPC peering supports bidirectional communication, but routing must be explicitly defined in both VPCs.
- It is best suited for scenarios where applications are split across multiple VPCs within the same provider.



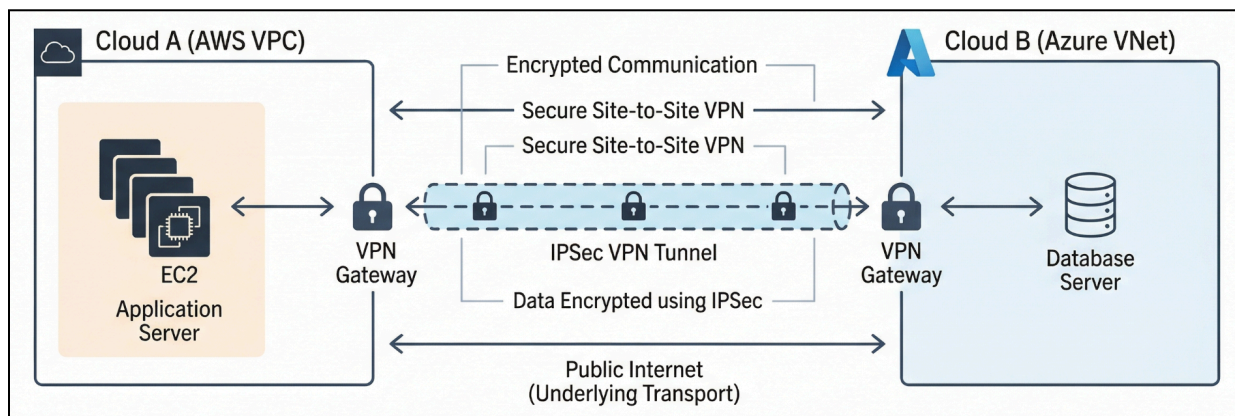
VPN Study:

A Virtual Private Network (VPN) is an encrypted connection over the public internet that allows secure communication between networks. It is commonly used for connecting on-premises systems to cloud or between different cloud providers.

- VPN uses IPSec (Internet Protocol Security) protocols to ensure data confidentiality, integrity, and authentication.
- It creates a secure tunnel through which data is transmitted safely over an otherwise insecure public network.
- VPN connections are widely used in hybrid cloud architectures to connect on-premises infrastructure with cloud resources.

Types of VPN:

- Site-to-Site VPN: Connects two networks (e.g., office network to cloud or cloud-to-cloud communication).
- Client VPN: Connects individual users or devices securely to a remote network.



Multi-Cloud Analysis:

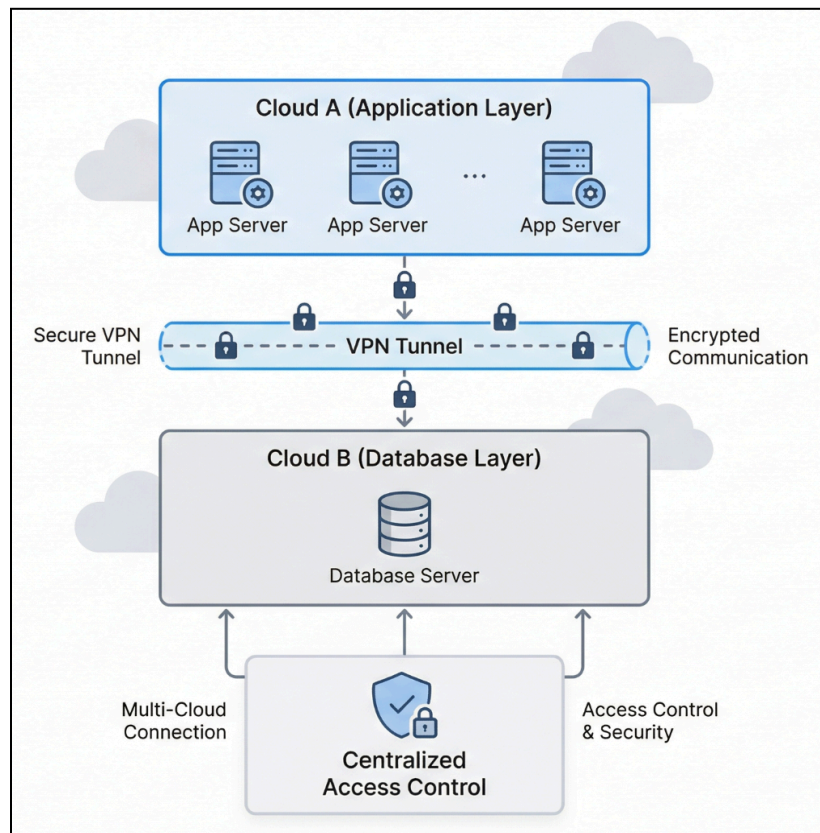
Connectivity Type	Scenario
VPC Peering	Same cloud provider
Site-to-Site VPN	AWS ↔ Azure / GCP
Dedicated Interconnect	High-performance enterprise
SD-WAN	Large distributed systems

VPC vs VPN Peering:

Based on the evaluation, a multi-cloud approach using AWS and Microsoft Azure is recommended. AWS can be used for its global infrastructure and scalability, while Azure can be leveraged for enterprise integration and compliance support. This strategy ensures high reliability, regulatory compliance, and reduced dependency on a single provider.

Parameter	VPC Peering	VPN
Encryption	Optional	Mandatory
Latency	Low	Medium
Cost	Lower	Higher
Multi-Cloud	No	Yes
Setup Complexity	Low	Medium

Multi-Cloud Architecture Design:



This architecture demonstrates a multi-cloud setup where the application is hosted in one cloud and the database in another. A secure VPN tunnel ensures encrypted communication between the two environments. Centralized access control enhances security and management.

Conclusion:

VPN and VPC peering concepts were studied and compared across multiple cloud environments. The experiment demonstrated how secure networking enables hybrid and multi-cloud architectures while balancing performance, cost, and security.

This experiment highlights the importance of secure networking in multi-cloud environments. VPNs and VPC peering enable controlled communication between isolated networks, forming the backbone of modern hybrid and multi-cloud architectures.